Examiner only

2. The total stopping distance for a moving car is given by the equation below:

total stopping distance = thinking distance + braking distance

(a) These distances may be affected by a number of factors.
Three of these factors are given in the table below.

Put a **tick** (✓) or a **cross** (**X**) in each box below to show whether the distance is affected by each factor. [3]

The first row has been done for you.

Factor	Thinking distance	Braking distance	Total stopping distance
Worn tyres	**X**	✓	✓
Drunk driver			
Wet road			

(b) At a speed of 13 m/s, the thinking distance of an alert driver is 9.1 m.

Use the equation:

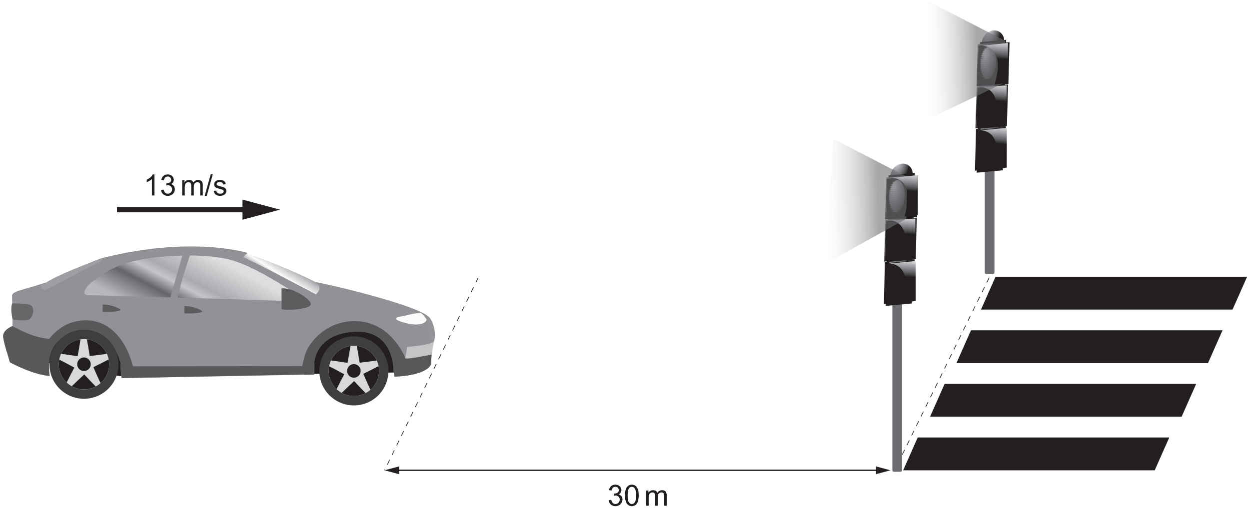
$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

to calculate the thinking time. [2]

Thinking time = ... s

Examiner only

(c) A driver of a car travelling at 13 m/s sees traffic lights 30 m ahead when the lights turn to red.



The thinking distance = 9.1 m and the braking distance = 13.9 m at this speed.

Use the equation:

total stopping distance = thinking distance + braking distance

to explain whether the car would be able to stop before reaching the crossing. [2]

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